William Liu

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Education

University of Alberta June 2024
Bachelor of Science, Specialization in Computing Science, With Distinction GPA: 3.4/4.0 | Major GPA: 3.8/4.0

Experience

Zebra Technologies Montreal, QC
Technical Communication Specialist – Machine Vision June 2024 – November 2025

- Co-led a cross-team initiative to create standardized process checklists across **4** teams, reducing errors, improving workflow consistency, and streamlining production of **10** manuals each month.
- Authored technical documentation for Zebra Aurora Imaging Library and Aurora Design Assistant, working closely with software engineers to document APIs, OCR, deep learning, 3D imaging, and industrial communication modules.
- Refactored QA testing for help-file releases by eliminating redundant cases and consolidating **6** hours of tests into a single **1**-hour test performed **3** times per release cycle.
- Managed **5** physical and **4** virtual machines, and expanded Linux testing for help documentation, detecting **2%** platform-specific errors in early tests and increasing test-platform coverage by **50%**.
- Optimized the help-file web upload workflow, reducing team effort by **80%** and improving cross-team documentation delivery coordination.
- Worked closely with computer vision engineers, software developers, project managers, and subject matter experts to ensure the accuracy and completeness of technical documents.
- Contributed to multiple product release cycles by participating in code reviews, testing, deployment, and maintenance.

Zebra Technologies (Matrox) September 2022 – May 2023, February 2024 – June 2024
Machine Vision Technical Writer Intern Montreal, QC

Projects

QR Score | *Java, Android Studio, Firestore*

- Led a team of 5 to develop an Android app using Java, Firebase Firestore, and Android Studio that allows users to scan QR codes, earn points, and compete on leaderboards.
- Utilized Firebase Authentication to implement a streamlined, username-less, and password-less login process.
- Incorporated OpenStreetMaps API to display QR Codes in various geographic locations.
- Executed unit and UI testing using JUnit and Robotium to detect issues early, reducing the time required to fix bugs.
- Adhered to agile scrum principles and utilized Git/Github for code review.

TopSongify | *Python*

- Developed a Spotify playlist generator based on the analyzed user's listening activity, such as the user's short-, medium-, and long-term listening activity.
- Integrated Spotify's Web API using Python Requests to retrieve and analyze user listening history for automated playlist generation.

Personal Portfolio Website | *HTML, CSS, JavaScript*

- Designed and built a responsive personal portfolio website using HTML, CSS, and JavaScript.
- Utilized Figma for mockup creation and Canva to design a personalized logo.

Skills

Languages: C++, Python, C, Java, JavaScript, SQL, Assembly, HTML, CSS/SCSS, XML
Technologies: Git/Github, Android Studio, Firebase Firestore, Figma, oXygen XML Author, CorelDRAW, Jira, Confluence

Relevant Coursework

- Software Engineering
- Operating Systems
- Software Quality
- Computer Architecture
- Data Structures
- Algorithms
- Digital Logic Design
- Discrete Mathematics
- Linear Algebra I, II